

# Dimple Bhuta

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## PROFESSIONAL PROFILE

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Robotics Engineer with 10+ years of experience in autonomous robotics and computer vision, specializing in navigation, perception, state estimation, and sensor fusion. Experienced in developing UAV and ground robotic systems using ROS/ROS2, with PX4/ArduPilot for UAVs and HIL/SITL simulation for system validation. Research experience includes GPS-denied navigation, visual positioning, path planning, obstacle avoidance, multispectral sensing, and multi-agent robotics. Skilled in algorithm development, hardware integration, simulation, field testing, and deployment.

## EDUCATION

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### Virginia Commonwealth University

Richmond, VA, USA

*Master of Science in Biomedical Engineering, GPA: 3.828/4.000 (Top 12% of class)*

*Aug. 2009 – May 2012*

- Thesis: Developed a standalone EEG-based embedded interface for locked-in patients, enabling real-time control of room lighting through intent detection using biomedical signal processing, embedded systems, and pattern recognition techniques.

### K. J. Somaiya College of Engineering, Mumbai University

Mumbai, India

*Bachelor of Engineering in Electronics*

*Aug. 2005 – May 2009*

- Final Year Project: Designed and implemented a secure cryptographic architecture integrating DES with a 64-bit Fibonacci LFSR in VHDL.

## PUBLICATIONS

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### Published Conference Publications

- D. Bhuta, M. S. Hashmi, S. Bhegde, Z. Shaikh, H. Arya, and A. Sinha, “Advancing Low-Altitude Agricultural UAV Operations: Plug-and-Play Device for Sensing and Obstacle Avoidance,” *IFAC-PapersOnLine*, vol. 59, no. 23, pp. 168–173, 2025.

### Accepted Papers

- D. Bhuta, A. Siyote, S. Bhegde, R. Kushwaha, S. Banerjee, and H. Arya, “UAV-Based Multispectral Imaging for Surface Oil and Water Leak Detection Over Soil Using Deep Learning,” accepted at *IEEE IGARSS 2026*.
- A. Pathak, D. Bhuta, S. Wayal, R. Kushwaha, S. Bhegde, A. G., and S. Gawande, “UAV Multispectral Time-Series and Weather-Based Hurdle Modeling for Onion Diseases and Pest Estimation,” accepted at *IEEE IGARSS 2026*.

## WORK EXPERIENCE

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### Project Manager

Sept. 2025 – Present

*Drone Center, IIT Bombay*

*Mumbai, India*

- \* Conducting research and developing GPS-denied terrain-following navigation algorithms for multirotor UAVs using point LiDAR, monocular vision, and 9-axis IMU data, leveraging VPS, VIO, and related localization techniques.
- \* Developing full-coverage UAV obstacle avoidance via multi-sensor placement (LiDAR, radar, depth cameras), with active research into multi-modal sensor fusion and high-speed navigation.
- \* Evaluated state-of-the-art object detection models for automated bunker detection.
- \* **Technologies:** ROS2, PX4, ArduPilot, Gazebo, Radar, RealSense D455, Sensor Fusion, PyTorch, NVIDIA Jetson

### Principal Engineer – State Estimation and Control

Apr. 2024 – Sept. 2025

### Senior Engineer – State Estimation and Control

June 2022 – Mar. 2024

*TIH Foundation for IoT & IoE, IIT Bombay*

*Mumbai, India*

- \* Led the development and field validation of lightweight sense-and-avoid systems for agricultural UAVs.
- \* Developed a multimodal UAV multispectral sensing and weather-station system for onion disease detection and estimation, aimed at providing precision agriculture advisory support to farmers.
- \* Implemented deep learning models for surface oil-water pipeline leakage detection using aerial multispectral imagery.
- \* Worked on UAV localization and navigation using visual-inertial odometry (VIO) systems.
- \* Developed odometry and localization pipelines for autonomous ground robotic platforms.
- \* **Technologies:** ROS, MAVROS, Gazebo, SITL, HITL, NVIDIA Jetson, PX4, Python, Linux

## Senior Research Fellow

Mar. 2019 – Aug. 2021

*Autonomous Robots and Multi-Robot Systems (ARMS) Lab, IIT Bombay*

*Mumbai, India*

- \* Worked on autonomous multi-agent robotic patrolling systems for surveillance and defense applications in collaboration with Centre for Artificial Intelligence and Robotics, Defence Research and Development Organisation, India.
- \* Developed trajectory generation and tracking pipelines for mobile robots.
- \* Implemented static and dynamic obstacle avoidance algorithms.
- \* Integrated robotic algorithms with ROS and Webots simulation environments.
- \* **Technologies:** ROS, Webots, MATLAB, Python

## Software Developer

July 2018 – Jan. 2019

*AISIGHT Video Analytics Pvt. Ltd.*

*Mumbai, India*

- \* Developed deep learning and computer vision algorithms for surveillance and anomaly detection systems.
- \* Implemented human detection and tracking pipelines for video analytics applications.
- \* **Technologies:** OpenCV, PyTorch, TensorFlow, C++, Python

## Drone and Robotics Engineer

Dec. 2017 – June 2018

*Dhristi Works*

*Mumbai, India*

- \* Developed motion planning algorithms for robotic manipulators.
- \* Implemented object detection, point-cloud processing, and grasping pipelines using ROS and MoveIt.
- \* **Technologies:** ROS, Gazebo, MoveIt, Python

## Research Assistant

Mar. 2016 – Mar. 2017

*Singapore Institute of Neurotechnology, National University of Singapore*

*Singapore*

- \* Developed robotic slip-control and haptic feedback systems using robotic manipulators and prosthetic interfaces.
- \* Worked on robotic manipulation and motion planning using UR10 robotic systems.
- \* **Technologies:** MATLAB, Simulink, C++, Python

## Project Assistant

Aug. 2015 – Dec. 2015

*Multimodal Perception Laboratory, IIIT Bangalore*

*Bangalore, India*

- \* Evaluated and implemented image segmentation algorithms for jewelry segmentation using OpenCV.
- \* Modified the GrabCut algorithm to improve segmentation performance.
- \* **Technologies:** C++, OpenCV

## Assistant Professor

Aug. 2014 – Apr. 2015

*Electronics and Communication Engineering, Haryana College of Technology and Management*

*Haryana, India*

- \* Taught courses on Digital Signal Processing and Microprocessors with Interfacing.
- \* Conducted laboratory sessions on Digital Communication and Microprocessors.

## Project Manager – Embedded Software Engineer

Sept. 2012 – Apr. 2013

*Infinite Biomedical Technologies*

*Baltimore, USA*

- \* Developed a pattern-recognition-based prosthetic arm control system using EMG signals for transradial amputees.
- \* Implemented EMG classification pipelines for hand-open, close, flex, extend, pronate, supinate, and hook-hand gestures.
- \* **Technologies:** C++, MATLAB

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## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++

**Robotics Frameworks and Middleware:** ROS, ROS2, MAVROS, PX4, ArduPilot

**Navigation and Estimation:** Sensor Fusion, State Estimation, Visual-Inertial Localization, Path Planning, Trajectory Tracking, Obstacle Avoidance

**Simulation and Validation:** Gazebo, Webots, MATLAB, Simulink, SITL, HITL

**Perception and AI:** OpenCV, PyTorch, Deep Learning, Object Detection

**Embedded Platforms and Sensors:** NVIDIA Jetson (Nano, Orin Nano, AGX Xavier), Raspberry Pi, STM32, Pixhawk, IMU, GNSS, Radar, LiDAR, Depth Cameras

**Tools and Operating Systems:** Linux, Git, MicroPython